Attention Is All You Need: A Novel Transformer Architecture for Sequence Modeling

Abstract: We propose the Transformer, a novel architecture based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Unlike recurrent models that process tokens sequentially, the Transformer processes all positions in parallel and computes relationships between any two positions with a constant number of operations.

Introduction: Sequence transduction models have traditionally relied on recurrent or convolutional networks. However, their sequential nature inhibits parallelization and makes long-range dependencies difficult to capture. The core innovation of our work is replacing the recurrence entirely with multi-head self-attention.

Method: The Transformer uses an encoder-decoder structure. Each layer applies multi-head self-attention followed by position-wise feed-forward networks, with residual connections and layer normalization. We introduce sinusoidal positional encodings to inject order information. Scaled dot-product attention is computed as $\mathrm{softmax}(QK^T/\sqrt{d_k})V$.

Experiments: We train on the WMT 2014 English-to-German and English-to-French translation tasks. Our model achieves a BLEU score of 28.4 on En-De, surpassing the previous best results while requiring significantly less training time than recurrent baselines. We also evaluate on English constituency parsing with good results.

Conclusion: The Transformer is the first transduction model relying entirely on self-attention, and it can be trained significantly faster than architectures based on recurrent or convolutional layers. We hope this work encourages more attention-based models across the field.